



BAULKHAM HILLS HIGH SCHOOL

2014

YEAR 12 HALF-YEARLY

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions
- Start a new page for each question

Total marks – 70

Exam consists of 11 pages.

This paper consists of TWO sections.

Section 1 – Pages 4-6

Multiple Choice

Question 1-10 (10 marks)

Section 2 – Pages 7-10

Extended Response

Question 11- 14 (60 marks)

Standard integrals provided on page 11

Section I - 10 marks**Allow about 15 minutes for this section****Use the multiple choice answer sheet for question 1-10**

1. Which of the following shapes is the locus of the point P representing the complex number z moving in an Argand diagram such that

$$|z - 2i| + |z + 2i| = 6$$

- (A) A circle
- (B) A parabola
- (C) A hyperbola
- (D) An ellipse

2. What is the multiplicity of the root $x = -1$ of the equation $3x^5 - 5x^4 - 35x - 27 = 0$?

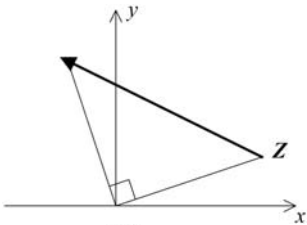
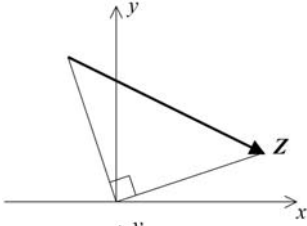
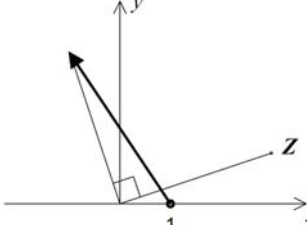
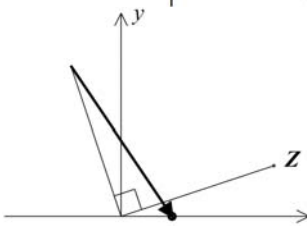
- (A) one
- (B) two
- (C) three
- (D) four

3. In modulus argument form $-\sqrt{2}(1 - i)$ is

- (A) $2 \left(\cos \left(\frac{3\pi}{4} \right) - i \sin \left(\frac{3\pi}{4} \right) \right)$
- (B) $2 \left(\cos \left(\frac{3\pi}{4} \right) + i \sin \left(\frac{3\pi}{4} \right) \right)$
- (C) $\cos \frac{\pi}{4} - i \sin \frac{\pi}{4}$
- (D) $-\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}$

4. If $y = \cos^{-1}(e^x)$, the expression for $\frac{dy}{dx}$ is

- (A) $-\operatorname{cosec} y$
- (B) $-\tan y$
- (C) $-\cot y$
- (D) $-\sec y$

5.	If α, β and γ are roots of the equation $x^3 + 5x^2 + 4 = 0$, then the cubic equation with roots α^2, β^2 and γ^2 is (A) $4x^3 + 5x^2 + 1 = 0$ (B) $x^3 - 25x^2 - 40x - 16 = 0$ (C) $x^3 + 25x^2 + 40x + 16 = 0$ (D) $x^3 + 25x^2 - 40x + 16 = 0$
6.	The equation of the normal to the hyperbola $x = 2 \sec \theta, y = \tan \theta$ at the point where $\theta = \frac{\pi}{4}$ is (A) $\sqrt{2}x + y - 5 = 0$ (B) $\sqrt{2}x + 2y - 2 = 0$ (C) $\sqrt{2}x - 2y - 2 = 0$ (D) $\sqrt{2}x - y + 5 = 0$
7.	On an Argand diagram, point Z is shown to represent the complex number z. Which diagram below shows the vector that represents $(1 - i)z$? (A)  (B)  (C)  (D) 

8.	Three of the six letters of the word R O M A R O are selected and arranged in a row. How many different arrangements are possible? (A) 120 (B) 30 (C) 42 (D) ${}^6P_3 \times 3!$
9.	Given the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, which of the following would be the equation of the asymptotes? (A) $y = \pm \frac{3}{4}x$ (B) $y = \pm \frac{4}{3}x$ (C) $y = \pm \frac{16}{9}x$ (D) $y = \pm \frac{16}{9}x$
10.	The locus of z in the Argand plane where $\arg(z - 2) - \arg(z - i) = \frac{\pi}{3}$ is (A) (B) (C) (D)
End of Section I	

Section II – Extended Response

Attempt questions 11-14. Show all necessary working.

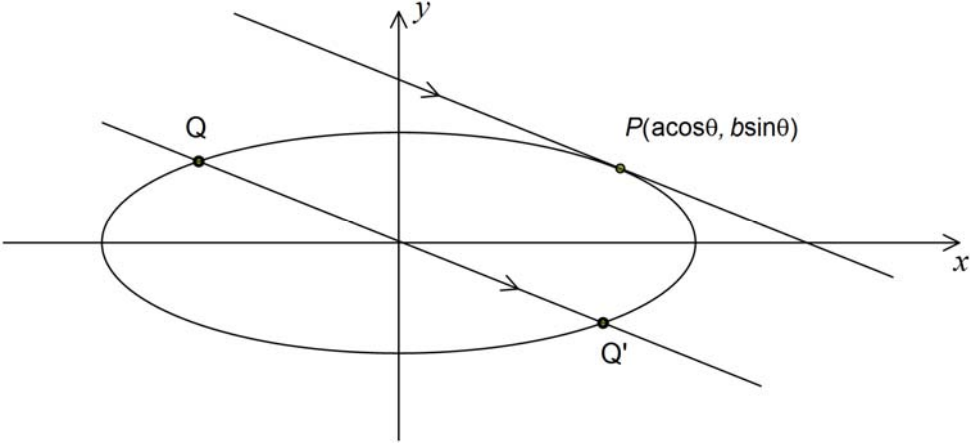
Answer each question on a SEPARATE PAGE Clearly indicate question number.

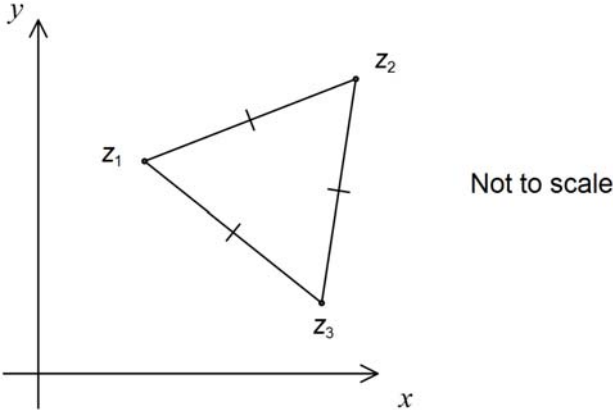
Each piece of paper must show your BOS number.

All necessary working should be shown in every question.

Question 11 (15 marks)		Marks
(a)	Let $z = i - \sqrt{3}$ and $w = 1 - i$, find	
	(i) $\bar{z} + w$	1
	(ii) zw in modulus /argument form.	3
(b)	(i) Find all the complex numbers $z = a + ib$ such that $(a + ib)^2 = 8 + 6i$, where a, b are real numbers.	2
	(ii) Hence solve $z^2 + 2z(1 + 2i) - (11 + 2i) = 0$	2
(c)	Solve $x^4 + 4x^3 - 16x - 16 = 0$ given that it has a root of multiplicity 3	3
(d)	Find the equation of the locus of all z , such that $ z - 2 = \operatorname{Re}(z)$.	2
(e)	For the hyperbola $xy = 12$, find its	
	(i) foci	1
	(ii) equations of directrices	1
End of Question 11		

Question 12 (15 marks)	Marks
a) A nine member committee consists of 4 male students, 3 female students and 2 teachers. The committee meets around a circular table so that the male students sit together as a group, and so do the female students, but no female student sits next to a male student. (i) How many different arrangements are possible? (ii) One particular male student does not wish to sit next to one particular teacher. How many ways can this be arranged?	2 2
b) The points represented by the complex number $z_1 = \sqrt{3} + i$ and two other complex numbers z_2 and z_3, lie on the circumference of a circle with centre O and radius 2. These three points are vertices of an equilateral triangle. Find the complex numbers z_2 and z_3 in the form $a + ib$ where a and b are real.	2
(c) Solve $z^4 - z^3 + 6z^2 - z + 15 = 0$ for z given that $z = 1 - 2i$ is a root of the equation.	2
(d) The locus of z is represented by $z - 1 = 1$ (i) Sketch the locus. (ii) Show that $z^2 - z = z$ (iii) Show that $\arg(z - 1) = \frac{2}{3} \arg(z^2 - z)$.	1 1 2
(e) The diagram shows points A, B and T marked on a circle. A tangent to the circle at T, KC is drawn such that BC is perpendicular to KC. TM is perpendicular to AB. Show that MC is parallel to AT.	3
End of Question 12	

Question 13 (15 marks)	Marks
a) Solve $ 3x^2 - 2x - 2 < 3x$	3
b) Given that ω is a non-real root of the equation $z^5 = 1$ and that $\alpha = \omega + \omega^4$ is a root of the quadratic equation $x^2 + bx + c = 0$, where b and c are real. (i) Prove that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$. 1 (ii) Find the second root β of the equation $x^2 + bx + c = 0$ in terms of positive powers of ω. 1 (iii) Find the values of the coefficients a and b. 2 (iv) Deduce that the exact value of $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ 2	
(c)  The point $P(a \cos \theta, b \sin \theta)$ lies on $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. A line is drawn through the origin parallel to the tangent at P. The line meets the ellipse at Q and Q'. (i) Show that the equation of the tangent at P is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ 2 (ii) Show that the equation of QQ' is $xb \cos \theta + ya \sin \theta = 0$ 1 (iii) Hence find the coordinates of Q and Q' 1 (iv) Prove that the area of the $\Delta QPQ'$ is independent of the position of P. 2	
End of Question 13	

Question 14 (15 marks)	Marks
a) Given that $\cos\theta + i\sin\theta \neq 1$, prove by mathematical induction that $1 + \operatorname{cis}\theta + \operatorname{cis}2\theta + \cdots + \operatorname{cis}(n\theta) = \frac{1 - \operatorname{cis}(n+1)\theta}{1 - \operatorname{cis}\theta}$ where $n \geq 0$, and $\operatorname{cis}\theta = \cos\theta + i\sin\theta$	3
b)  If complex numbers z_1, z_2, z_3 represent vertices of an equilateral triangle (i) Express $z_2 - z_1$ in terms of $z_3 - z_1$ (ii) Hence prove $z_1^2 + z_2^2 + z_3^2 = z_1 \cdot z_2 + z_2 \cdot z_3 + z_3 \cdot z_1$	1 3
c) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, where, $p \neq q$, lie on the rectangular hyperbola with equation $xy = c^2$. (i) Show that the equation of the tangent to the hyperbola $xy = c^2$ at the point $P\left(cp, \frac{c}{p}\right)$ is $x + p^2y = 2cp$ (ii) If the tangents at $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ meet at the point $R(x_0, y_0)$, prove that $p + q = \frac{2c}{y_0}$ and $pq = \frac{x_0}{y_0}$ (iii) If the length of chord PQ is d units, show that $d^2 = c^2(p - q)^2 \left\{1 + \frac{1}{p^2q^2}\right\}$ (iv) If d remains fixed, deduce that the locus of R has equation $4c^2(x^2 + y^2)(c^2 - xy) = x^2y^2d^2$	2 3 1 2
End of Question 14	

End of Paper

Yr. 12 Extension 2 - Half Yearly 2014 Solutions

Multiple choice

1. **D** ellipse

2. **B** $P(-1) = 0$
 $P'(x) = 15x^4 - 20x^3 - 35 \therefore P'(-1) = 0$
 $P''(x) = 60x^3 - 60x^2 \therefore P''(-1) \neq 0$ } $\therefore \text{multiplicity} = 2$

3. **A** $-\sqrt{2}(1-i) = -\sqrt{2} + \sqrt{2}i = z$

$$\text{modulus} = \sqrt{(-\sqrt{2})^2 + (\sqrt{2})^2} = 2 \quad \arg z = \tan^{-1}\left(\frac{-\sqrt{2}}{\sqrt{2}}\right) = \frac{3\pi}{4}$$

$$\therefore z = 2\left(\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right)$$

4. **A** $(2\sec\theta, \tan\theta)$ at $\theta = \frac{\pi}{4}$ $a=2$ $b=1$
 $= \left(2\sec\frac{\pi}{4}, \tan\frac{\pi}{4}\right) = (2\sqrt{2}, 1)$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \therefore \frac{x^2}{4} - y^2 = 1 \therefore \frac{2x}{4} - 2y \cdot \frac{dy}{dx} = 0$$

$$\therefore m_T = \frac{2\sqrt{2}}{4 \times 1} = \frac{\sqrt{2}}{2} \therefore m_N = -\frac{2}{\frac{\sqrt{2}}{2}} = -\sqrt{2} \therefore \frac{dy}{dx} = \frac{x}{4y}$$

$$\therefore \text{eqn. of normal} \therefore y-1 = -\sqrt{2}(x-2\sqrt{2}) \therefore \sqrt{2}x + y - 5 = 0$$

5. **B** let $y = x^2 \therefore x = \sqrt{y} \therefore P(x) = 0 \therefore P(\sqrt{y}) = 0$

$$\therefore (\sqrt{y})^3 + 5(\sqrt{y})^2 + 4 = 0$$

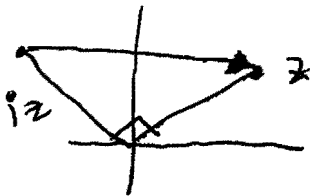
$$y\sqrt{y} = -5y-4 \therefore y^3 = (-5y-4)^2 = 25y^2 + 40y + 16$$

$$\therefore \text{cubic equation is } y^3 - 25y^2 - 40y + 16 = 0 \text{ or } x^3 - 25x^2 - 40x + 16 = 0$$

6. **C** $y = \cos^{-1}(e^x) \therefore e^x = \cos y$

$$\frac{dy}{dx} = \frac{-e^x}{\sqrt{1-(e^x)^2}} = \frac{-\cos y}{\sqrt{1-\cos^2 y}} = \frac{-\cos y}{\sin y} = -\cot y$$

7. **B** $(1-i)z = z - iz$



8. [C]

choose R, R and one of O, M, A $\therefore {}^3C_1 \times \frac{3!}{2!} = 9$

choose O, O, and one of R, M, A $\therefore 9$

arrange [R] [O] M, A $\therefore {}^4P_4 = 4! = 24$

$\therefore \text{total} = 9 + 9 + 24 = 42$

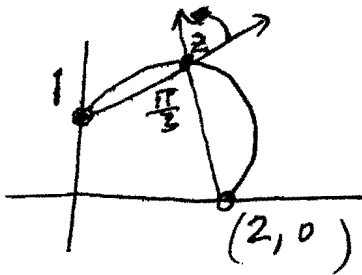
9. [B]

$$\frac{y^2}{16} = \frac{x^2}{9} - 1$$

$$\frac{y^2}{x^2} = \frac{16}{9} - \frac{16}{x^2} \therefore \text{asymptote } \frac{y}{x} = \pm \frac{4}{3}$$

$$\therefore y = \pm \frac{4}{3} x$$

10. [A] $\arg \frac{(z-2)}{(z-i)} = \frac{\pi}{3}$



Question 11

BOS#:

$$(a) \quad z = i - \sqrt{3}, \quad w = 1 - i$$

$$(i) \quad \bar{z} + w = -\sqrt{3} - i + 1 - i = (-\sqrt{3} + 1) - 2i \quad (1)$$

$$(ii) \quad z \cdot w = (-\sqrt{3} + i)(1 - i) = (-\sqrt{3} + 1) + i(1 + \sqrt{3}) \quad (1/3)$$

$$(OR) \quad |z \cdot w| = |-\sqrt{3} + i| |1 - i| = |2| \times |\sqrt{2}| = 2\sqrt{2} \quad (1)$$

$$\arg(z \cdot w) = \arg(z) + \arg(w) = \frac{5\pi}{6} + \left(-\frac{\pi}{4}\right) = \frac{7\pi}{12} \quad (1)$$

$$\therefore zw = 2\sqrt{2} \operatorname{cis} \frac{7\pi}{12} \quad (1)$$

$$(b)(i) \quad (a + ib)^2 = 8 + 6i$$

$$a^2 - b^2 + 2iab = 8 + 6i$$

$$(1) \quad a^2 - b^2 = 8 \quad (1)$$

$$(2) \quad 2ab = 6 \quad \therefore b = \frac{3}{a}$$

$$\therefore a^2 - \left(\frac{3}{a}\right)^2 = 8 \quad \therefore a^4 - 8a^2 - 9 = 0$$

$$(a^2 - 9)(a^2 + 1) = 0$$

$$a = \pm 3 \quad \text{no solution in real}$$

$$\therefore \begin{matrix} a=3 \\ b=1 \end{matrix} \quad \begin{matrix} a=-3 \\ b=-1 \end{matrix} \quad \therefore z = \pm (3 + i) \quad (1)$$

$$(ii) \quad z^2 + 2z(1 + 2i) - (11 + 2i) = 0$$

$$z = \frac{-2(1 + 2i) \pm \sqrt{4(1 + 2i)^2 - 4(11 + 2i)}}{2}$$

$$= \frac{-2(1 + 2i) \pm \sqrt{(1 + 2i)^2 + (11 + 2i)}}{2} \quad (1)$$

$$= -(1 + 2i) \pm \sqrt{8 + 6i} = -(1 + 2i) \pm (\pm 3 + i)$$

Question 11

BOS#:

11b) ii) cont. $\therefore z = 2 - i$ or $-4 - 3i$ ①

c) $x^4 + 4x^3 - 16x - 16 = 0$

let $P(x) = x^4 + 4x^3 - 16x - 16$

$P'(x) = 4x^3 + 12x^2 - 16 = 4(x^3 + 3x^2 - 4)$

$P''(x) = 12x^2 + 24x = 0 = 12x(x + 2)$

when $x = 0, x = -2$

but $P(0) \neq 0 \therefore 0$ is not a root ①

$P(-2) = 0, P'(-2) = 0 \times P''(-2) = 0$

$\therefore x = -2$ is a root multiplicity 3 ①

$\therefore (x^4 + 4x^3 - 16x - 16) \div (x + 2)^3 = x - 2$

$\therefore x = -2$ and $x = 2$ (single root)
multiplicity 3 ①

d) $|z - 2| = \operatorname{Re}(z)$ let $z = x + iy$

$\therefore |x + iy - 2| = x$

$\therefore \sqrt{(x-2)^2 + y^2} = x$ ①

$(x-2)^2 + y^2 = x^2$

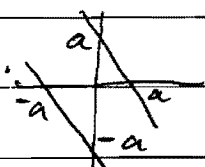
$x^2 - 4x + 4 + y^2 = x^2$

$\therefore y^2 = 4x - 4$ ① locus of z

e) $xy = 12 = c^2 = \frac{a^2}{2} \therefore 12 = \frac{a^2}{2} \therefore a = \sqrt{24} = 2\sqrt{6}$

$S, S'(\pm a, \pm a) \therefore S(+2\sqrt{6}, +2\sqrt{6}), S'(-2\sqrt{6}, -2\sqrt{6})$ ①

directrices:

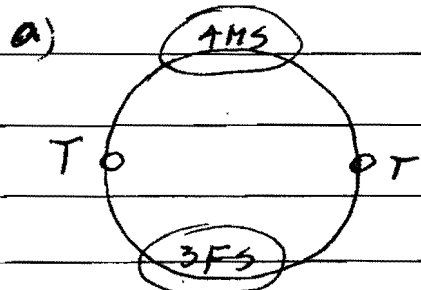


$\therefore x + y = 2\sqrt{6}$
or $x + y = -2\sqrt{6}$ ①

You may ask for extra writing paper if you need more space to answer question 11

Question 12

BOS#:

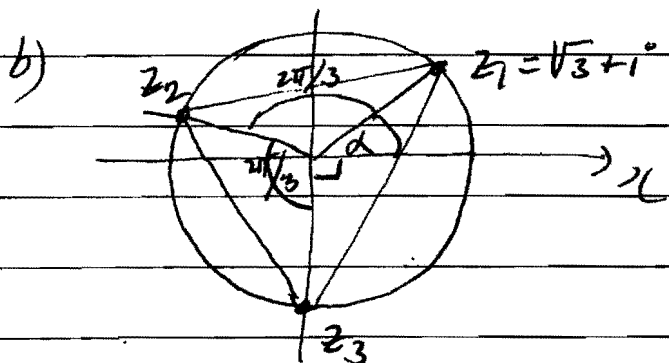


i) $4! \cdot 3! \cdot 2 = \underline{288 \text{ ways}}$ (1)

↓
teachers

ii) a particular male student M_1 sits next to a particular teacher T_1 in $3! \cdot 3! \cdot 2 = 72 \text{ ways}$ (1)

$\therefore$ number of ways M_1 doesn't sit next to $T_1 = 288 - 72 = \underline{216 \text{ ways}}$ (1)



$z_1 = \sqrt{3} + i = 2 \text{ cis } \frac{\pi}{6}$

$\alpha = \frac{\pi}{6}$

$\therefore z_2 = 2 \text{ cis } \left(\frac{\pi}{6} + \frac{2\pi}{3} \right)$

$= 2 \text{ cis } \frac{5\pi}{6}$

$\therefore z_2 = 2 \left(-\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = -\sqrt{3} + i$ (1)

$z_3 = z_2 \cdot \text{cis} \left(\frac{2\pi}{3} \right) = 2 \text{ cis } \frac{3\pi}{2} = 2 \text{ cis} \left(-\frac{\pi}{2} \right) = -2i$ (1)

c) ω is a root of $z^5 = 1 \therefore \omega^5 = 1$ or $\omega^5 - 1 = 0$

but $\omega^5 - 1 = (\omega - 1)(\omega^4 + \omega^3 + \omega^2 + \omega + 1) = 0$

Since ω non-real $\therefore \omega \neq 1 \therefore \omega^4 + \omega^3 + \omega^2 + \omega + 1 = 0$

Now quadratic equation $x^2 + bx + c = 0$

where $\alpha = \omega + \omega^4$, $\beta = \omega^2 + \omega^3$. But $\alpha + \beta = -b$ (1)

$\therefore \omega + \omega^4 + \omega^2 + \omega^3 = -b = -1$

$\therefore b = 1$

Question 12

BOS#:

$$12c) \text{ cont: } d \cdot \beta = \frac{c}{a}$$

$$\therefore (c\omega + c\omega^4)(c\omega^2 + c\omega^3) = c\omega^3 + c\omega^4 + c\omega^6 + c\omega^7 = c$$

$$\text{since } \left. \begin{array}{l} c\omega^6 = c\omega^5 \cdot c\omega = c\omega \\ c\omega^7 = c\omega^5 \cdot c\omega^2 = c\omega \end{array} \right\} \therefore \underbrace{c\omega^3 + c\omega^4 + c\omega + c\omega^2}_{-1} = c = -1 \quad (1)$$

$$\therefore \text{quad. equation is } x^2 + x - 1 = 0$$

$$d) \text{ solve } z^4 - z^3 + 6z^2 - z + 15 = 0 \text{ if } z = 1 - 2i \text{ is root}$$

$$\therefore \text{coeff. are real } \therefore 1 + 2i^0 = \bar{z} \text{ is root also}$$

$$\therefore \text{quadratic factor is } [z - (1 - 2i)][z - (1 + 2i)] = z^2 - 2z + 5 \quad (1)$$

$$\text{now } (z^4 - z^3 + 6z^2 - z + 15) \div (z^2 - 2z + 5) = z^2 + z + 3$$

$$-(z^4 - 2z^3 + 5z^2)$$

$$z^3 + z^2 - z + 15$$

$$-(z^3 - 2z^2 + 5z)$$

$$3z^2 - 6z + 15$$

$$-(3z^2 - 6z + 15)$$

$$0$$

$$\therefore \text{the other roots are solutions of } z^2 + z + 3 = 0$$

$$\therefore z = \frac{-1 \pm \sqrt{11}}{2} \quad (1)$$

$$\therefore \text{roots are: } 1 - 2i, 1 + 2i, -\frac{1}{2} + \frac{\sqrt{11}}{2}, -\frac{1}{2} - \frac{\sqrt{11}}{2}$$

$$e) 3x^2 - 2x - 2 < |3x|$$

$$(x > 0) \therefore 3x^2 - 2x - 2 < 3x \text{ or } (x < 0) \therefore 3x^2 - 2x - 2 < -3x$$

$$3x^2 - 5x - 2 < 0$$

$$(3x + 1)(x - 2) < 0$$

$$\rightarrow -\frac{1}{3} < x < 2$$

$$3x^2 + x - 2 < 0$$

$$(3x - 2)(x + 1) < 0$$

$$\rightarrow -1 < x < \frac{2}{3}$$

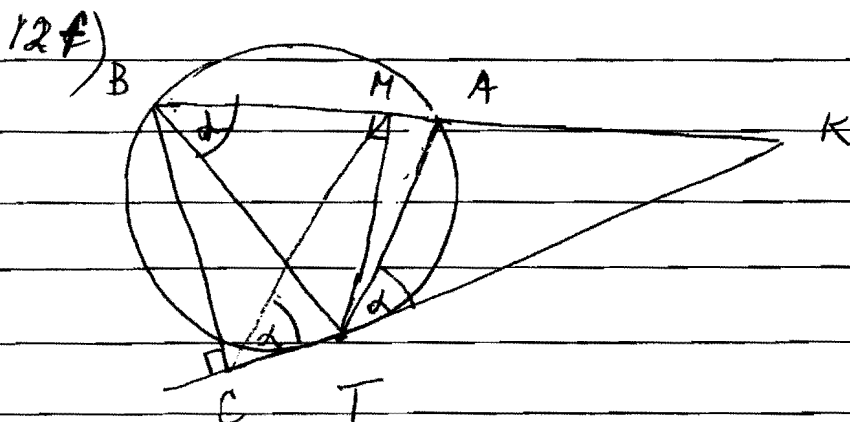
$$\therefore 0 < x < 2$$

solutions

$$\therefore -1 < x < 0$$

You may ask for extra writing paper if you need more space to answer question 12

Question 12

BOS#:

CBMT is a cyclic quadrilateral $\rightarrow \angle BMT = \angle BCT = 90^\circ$
 $\therefore$ Opposite $\angle$ s in quadrilateral

① $\therefore$ opposite $\angle$'s in quadrilateral are supplementary

Now $\angle ATK = \alpha = \angle TBM$ ($\angle$ between tangent & chord at point of contact = $\angle$ in alternate segment subtended by that chord)

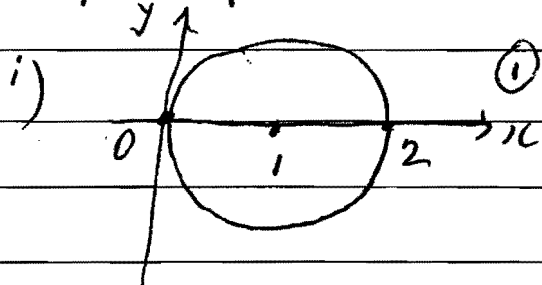
But $\angle MCT = \angle MBT = x$ (angles standing on the same arc TM of circle $CBMT$)

$\therefore AT \parallel MC$ (since $\angle ATK = \angle MCT$ and they're corresponding $\angle$'s) ①

Question 13

BOS#:

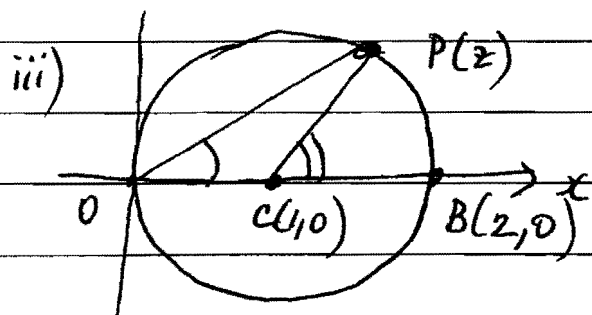
a) $|z-1|=1$



Locus is a circle

centre $(1, 0)$ radius $r=1$

ii) $|z^2 - z| = |z(z-1)| = |z| |z-1| = |z| \times 1 = |z| \therefore \text{shows}$



Let P be the pt. representing z
centre $C(1, 0)$, pt. $B(2, 0)$

Now $\angle PCB = 2 \times \angle POC$ (angle at the centre = double angle at circumference)

but $\angle POC = \arg z$ and $\angle PCB = \arg(z-1)$

$\therefore \arg(z-1) = 2 \cdot \arg z$ ①

Now $\frac{2}{3} \arg(z^2 - z) = \frac{2}{3} \arg[z(z-1)] = \frac{2}{3} [\arg z + \arg(z-1)]$
① $= \frac{2}{3} [\arg z + 2 \cdot \arg z] = \frac{2}{3} (3 \arg z) = 2 \arg z = \arg(z-1)$

b) i) since $z^5 - 1 = (z-1)(z^4 + z^3 + z^2 + z + 1)$

$\therefore$ roots of $z^4 + z^3 + z^2 + z + 1 = 0$ are amongst roots of $z^5 - 1 = 0$, where $z \neq 1$. ①

$\therefore$ roots are $z_1 = \text{cis } \frac{2\pi}{5}$, $z_2 = \text{cis } \frac{4\pi}{5}$

$z_3 = \bar{z}_2 = \text{cis } \left(-\frac{4\pi}{5}\right)$

$z_4 = \bar{z}_1 = \text{cis } \left(-\frac{2\pi}{5}\right)$ ①

Question 13

BOS#:

$$\begin{aligned}
 \text{13b) ii) Now } z^4 + z^3 + z^2 + z + 1 &= [(z - z_1)(z - z_2)(z - z_3)(z - z_4)] \\
 &= (z - z_1)(z - z_2)(z - \bar{z}_2)(z - \bar{z}_1) \\
 &= (z^2 - 2z \cdot \operatorname{Re} z_1 + 1)(z^2 - 2z \cdot \operatorname{Re} z_2 + 1)
 \end{aligned}$$

$$\therefore z^4 + z^3 + z^2 + z + 1 = \left[z^2 - 2z \cdot \cos \frac{2\pi}{5} + 1 \right] \left[z^2 - 2z \cdot \cos \frac{4\pi}{5} + 1 \right]$$

Now match up coefficients of z

$$\therefore z = -2z \cdot \cos \frac{2\pi}{5} + (-2z \cdot \cos \frac{4\pi}{5})$$

$$\therefore 1 = -2 \left(\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} \right)$$

$$\therefore -1/2 = \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} \quad \therefore \text{proven}$$

$$\text{c) i) } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \therefore \frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0 \quad \therefore \frac{dy}{dx} = -\frac{b^2}{a^2} \cdot \frac{x}{y}$$

$$\therefore \text{at } P(a \cos \theta, b \sin \theta) \quad m_T = -\frac{b^2}{a^2} \cdot \frac{a \cos \theta}{b \sin \theta} = -\frac{b \cos \theta}{a \sin \theta} \quad (1)$$

$$\therefore \text{equation of tangent: } y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$\frac{y \sin \theta}{b} - \sin^2 \theta = \frac{-\cos \theta}{a} x + \cos^2 \theta \quad (1)$$

$$\therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = \cos^2 \theta + \sin^2 \theta = 1 \quad \therefore \text{shown}$$

$$\text{ii) } QQ' \parallel \text{tangent at } P \quad \therefore m_{QQ'} = -\frac{b \cos \theta}{a \sin \theta}$$

and $(0, 0) \in QQ'$: equation

$$\text{of } QQ' : y = -\frac{b \cos \theta}{a \sin \theta} x \quad (1)$$

$$\therefore y a \sin \theta = -b \cos \theta x$$

$$\therefore QQ' \text{ is : } x b \cos \theta + y a \sin \theta = 0$$

You may ask for extra writing paper if you need more space to answer question 13

Question 13

BOS#:

$$13c) \text{ iii) } \begin{cases} xb \cos \theta + ya \sin \theta = 0 & (1) \\ Q, Q' \left\{ \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 & (2) \right. \end{cases}$$

$$(1) x = \frac{-ya \sin \theta}{b \cos \theta} \therefore (2) \frac{\frac{y^2 a^2 \sin^2 \theta}{b^2 \cos^2 \theta}}{a^2} + \frac{y^2}{b^2} = 1$$

$$\therefore \frac{y^2 \sin^2 \theta}{b^2 \cos^2 \theta} + \frac{y^2}{b^2} = 1 \therefore \frac{y^2}{b^2} \left[\frac{\sin^2 \theta}{\cos^2 \theta} + 1 \right] = 1$$

$$\therefore y^2 \left[\frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta} \right] = b^2 \quad (1) \therefore y^2 = b^2 \cos^2 \theta \therefore y = \pm b \cos \theta$$

$$\therefore x_{Q, Q'} = -y \frac{a \sin \theta}{b \cos \theta} = \begin{cases} -(b \cos \theta) \frac{a \sin \theta}{b \cos \theta} = -a \sin \theta \\ -(-b \cos \theta) \frac{a \sin \theta}{b \cos \theta} = +a \sin \theta \end{cases}$$

$$\therefore Q(-a \sin \theta, +b \cos \theta) \quad Q'(a \sin \theta, -b \cos \theta) \quad (1)$$

$$\text{iv) Area}_{\Delta QPQ'} = \frac{1}{2} \cdot QQ' \times h$$

$$QQ' = \sqrt{(2a \sin \theta)^2 + (2b \cos \theta)^2} = 2 \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$h = \text{perp. distance of } QQ' \text{ to } xb \cos \theta + ya \sin \theta = 0 \text{ from } P(a \cos \theta, b \sin \theta)$

$$\therefore h = \frac{|a \cos \theta \cdot b \cos \theta + b \sin \theta \cdot a \sin \theta|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \stackrel{(1)}{=} \frac{|ab| \cdot |\cos^2 \theta + \sin^2 \theta|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$\therefore \text{Area}_{\Delta QPQ'} = \frac{1}{2} \frac{|ab|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \cdot 2 \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta} = \underbrace{|ab|}_{\text{constant}}$$

which is independent of position of P.

Question 14

BOS#:

a) ^{STEP 1} Let $n=0$ $\therefore$ LHS = 1

$$RHS = \frac{1 - \cos \theta}{1 - \cos \theta} = 1$$

$\therefore$ the statement is true for $n=0$.

STEP 2: Assume that statement is true for $n=k$

$$\text{i.e. } 1 + \cos \theta + \cos 2\theta + \dots + \cos k\theta = \frac{1 - \cos(k+1)\theta}{1 - \cos \theta}$$

STEP 3: Now prove statement true for $n=k+1$ ie. $1 + \cos \theta + \dots + \cos(k+1)\theta = \frac{1 - \cos(k+2)\theta}{1 - \cos \theta}$

$$LHS = \underbrace{1 + \cos \theta + \dots + \cos k\theta}_{\text{assumption}} + \cos(k+1)\theta = \frac{1 - \cos(k+1)\theta}{1 - \cos \theta} + \cos(k+1)\theta$$

$$= \frac{1 - \cos(k+1)\theta + (1 - \cos \theta)(\cos(k+1)\theta)}{1 - \cos \theta}$$

$$= \frac{1 - \cos(k+1)\theta + \cos(k+1)\theta - \cos \theta \cdot \cos(k+1)\theta}{1 - \cos \theta}$$

$$= \frac{1 - \cos[\theta + (k+1)\theta]}{1 - \cos \theta} = \frac{1 - \cos(k+2)\theta}{1 - \cos \theta} = RHS$$

$\therefore$ the statement is true for $n=k+1$ if it's true for $n=k$, and since proven true for $n=0$
 $\therefore$ by math. induction the statement is true for all integers $n \geq 0$.

You may ask for extra writing paper if you need more space to answer question 15

Question 14

BOS#:

$$b) i) z_2 - z_1 = (z_3 - z_1) \operatorname{cis} \frac{\pi}{3} \quad (1)$$

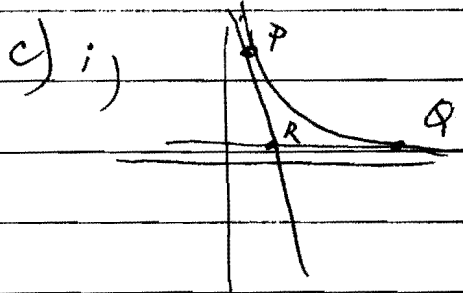
$$ii) z_1 - z_3 = (z_2 - z_3) \operatorname{cis} \frac{\pi}{3}$$

$$\therefore \frac{z_2 - z_1}{z_1 - z_3} = \frac{(z_3 - z_1) \operatorname{cis} \frac{\pi}{3}}{(z_2 - z_3) \operatorname{cis} \frac{\pi}{3}} \quad (1)$$

$$\therefore (z_2 - z_1)(z_2 - z_3) = (z_3 - z_1)(z_1 - z_3) \quad (1)$$

$$(z_2^2 - z_1 z_2 + z_2 z_3 - z_1 z_3) = (z_3 z_1 - z_1 z_3 - z_3^2 + z_1 z_3)$$

$$\therefore z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_1 z_3 \therefore \text{proven}$$



$$xy = c^2 \therefore y = \frac{c^2}{x}$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

$$\text{at } P(cp, \frac{c}{p}) \therefore m_T = -\frac{c^2}{c^2 p^2} = -\frac{1}{p^2} \quad (1)$$

eqn. of tangent at P is

$$y - \frac{c}{p} = -\frac{1}{p^2} (x - cp) \quad (1)$$

$$p^2(y - \frac{c}{p}) = -(x - cp)$$

$$p^2 y - cp = -x + cp$$

$$\therefore x + p^2 y = 2cp \quad (i) \text{ proven}$$

$$ii) \text{ similarly at } Q(cq, \frac{c}{q}) \text{ tangent is } \therefore x + q^2 y = 2cq \quad (ii)$$

$$R(x_0, y_0) \begin{cases} x_0 + p^2 y_0 = 2cp & (i) \\ x_0 + q^2 y_0 = 2cq & (ii) \end{cases}$$

$$p^2 y_0 - q^2 y_0 = 2cp - 2cq \quad (1)$$

$$(p^2 - q^2) y_0 = 2c(p - q)$$

$$(p - q)(p + q) y_0 = 2c(p - q) \quad | \div (p - q)$$

$$\text{if } |p| \neq |q| \therefore p - q \neq 0 \therefore (p + q) y_0 = 2c$$

$$\therefore p + q = \frac{2c}{y_0} \quad (1)$$

Question 14

BOS#:

14c) cont.

From (i) $x_0 = 2cp - p^2 y_0 \quad / \div y_0$

$$\therefore \frac{x_0}{y_0} = \frac{2cp}{y_0} - \frac{p^2 y_0}{y_0} \quad (1) \text{ but } \left(\frac{2c}{y_0} \right) = p + q$$

$$\therefore \frac{x_0}{y_0} = (p+q) \cdot p - p^2 = p^2 + qp - p^2 = pq$$

$$\therefore \frac{x_0}{y_0} = pq \quad (\text{proven})$$

iii) $PQ^2 = d^2 \quad (\text{since } PQ = d)$

$$\begin{aligned} \therefore d^2 &= (cp - cq)^2 + \left(\frac{c}{p} - \frac{c}{q} \right)^2 \\ &= c^2 (p-q)^2 + c^2 \left(\frac{1}{p} - \frac{1}{q} \right)^2 \\ &= c^2 \left[(p-q)^2 + \frac{(q-p)^2}{p^2 q^2} \right] \quad (1) \quad [(p-q)^2 = (q-p)^2] \\ &= c^2 (p-q)^2 \left[1 + \frac{1}{p^2 q^2} \right] \end{aligned}$$

iv) but $(p-q)^2 = (p+q)^2 - 4pq$
sub. this into (iii)

$$\therefore d^2 = c^2 \left[(p+q)^2 - 4pq \right] \left[1 + \frac{1}{p^2 q^2} \right]$$

also $(p+q) = \frac{2c}{y_0}$ and $pq = \frac{x_0}{y_0}$

$$\therefore d^2 = c^2 \left[\left(\frac{2c}{y_0} \right)^2 - 4 \left(\frac{x_0}{y_0} \right) \right] \left[1 + \frac{1}{\left(\frac{x_0}{y_0} \right)^2} \right]$$

$$d^2 = c^2 \left[\frac{4c^2}{y_0^2} - \frac{4x_0}{y_0} \right] \left[1 + \frac{y_0^2}{x_0^2} \right]$$

$$d^2 = \frac{4c^2}{y_0^2} (c^2 - x_0 y_0) \left[\frac{x_0^2 + y_0^2}{x_0^2} \right]$$

Question 14

BOS#:

14 c) iv) cont.

$$\therefore d^2 x_0^2 y_0^2 = 4c^2 (c^2 - x_0 y_0) (x_0^2 + y_0^2)$$

but $R(x_0, y_0) \in$ on the locus

$$\therefore \text{locus of } R(x_0, y_0) \text{ is: } d^2 x y^2 = 4c^2 (c^2 - xy) (x^2 + y^2)$$

$$\therefore 4c^2 (x^2 + y^2) (c^2 - xy) = x^2 y^2 d^2.$$

You may ask for extra writing paper if you need more space to answer question 16